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# Influence maximization in social networks: Theories, methods and challenges

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ABSTRACT

Influence maximization (IM) is the process of choosing a set of seeds from a social network so that the most individuals will be influenced by them. Calculating the social effect of a given seed set and identifying the smallest seed set that maximizes the social influence present two theoretical challenges in the IM problem, so the majority of current research projects focus on finding solutions to these two problems. This paper presents a comparative study of advanced approaches to IM algorithms, with particular attention to the following key aspects: (1) To establish the foundation of the IM problem, three types of diffusion models are first reviewed and their features are compared. Based on the classical diffusion models, research works on context-aware diffusion models and deep learning-based simulation of diffusion processes are summarized; (2) a complete analysis and comparison of classic IM algorithms and context-aware IM algorithms are presented using the algorithmic framework of IM algorithms; and (3) key challenges and future research directions in this field are explored. For researchers who are new to the IM field, our study offers the most recent advancements in the sector, a deeper grasp of IM issues, and a solid foundation from which to pursue further work in the area.

## 1. Introduction

Maximizing influence in social networks can be explained in a broader sense. To begin, consider a social network as a directed graph with users as nodes and their connections to one another as directed edges. This product is promoted throughout the social network graph via the word-of-mouth effect, which is the effect of human-to-human transmission. Promotion can result in either rapid decline or rapid growth. These two outcomes depend on three key questions. The first question is how to choose the people who are initially given the right to try the product. This is the seed node set S mentioned later. The second question is, how does this small group of people get other people to be influenced to buy the product? In fact, this process is the capability of different diffusion models. The last question is how to estimate the number of users affected by this small group of people, i.e., the most influential people. This is actually the problem of estimating the propagation influence of a given seed set. According to the three problems mentioned above, the influence maximization problem in social networks can be defined as finding the seed set composed of the smallest number of nodes in the social network so that they can influence the largest number of users in the social network; that is, influence maximization.

Influence maximization is most commonly used in social network viral marketing; that is, identifying potential customers for marketing purposes, with the goal of minimizing marketing costs and maximizing profits. Using the 'word of mouth' effect, viral marketing tries to find a small group of well-known people to try out a product before getting more people to use it [1,2].

Influence maximization was first considered as an algorithmic problem by [3,4], who employed Markov fields to address the IM problem and probabilistic models to define it. However, it is still unclear how users communicate with one another and how the diffusion process will continue after a certain group of users has been chosen. So Kempe et al. [5] defined the problem for the first time as a discrete optimization problem, specified two diffusion models, and theoretically demonstrated that the IM problem is NP-hard under these two diffusion models.

Finally, a greedy approach that can approximate the optimal solution with an approximation ratio of $1 - 1/e$ is proposed. It makes use of the influence function's submodularity. They also showed experimentally that their greedy algorithm significantly outperforms the classical degree and centrality-based heuristics in terms of influence propagation. However, this simulation-based approach [5] is very time-consuming, so many optimization methods based on greedy algorithms have emerged. Since then, the influence maximization problem has jumped out of the marketing or business domain and focused on problem formulation and model learning.

This review aims to provide a comprehensive study of existing advanced IM algorithms from an algorithmic perspective, focusing on four areas as shown in Fig. 1.

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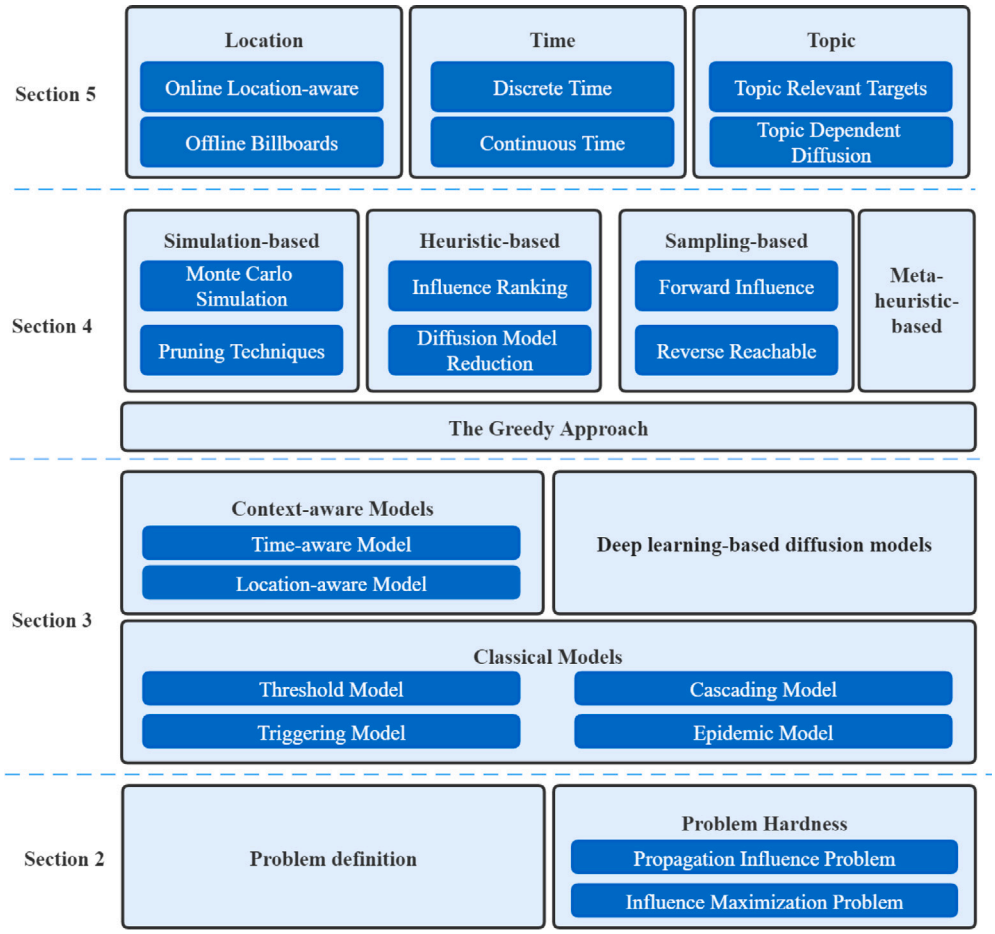


**Fig. 1.** The survey overview.

- IM Problem (Section 2). The IM problem is formally defined from the mathematical level, and the two main theoretical difficulties of estimating the diffusion impact and maximizing the impact under the diffusion model are introduced in more detail.
- Diffusion Models (Section 3). As an important part of the influence maximization problem, three types diffusion models have been proposed, namely classical diffusion models, context-aware models, and deep learning-based diffusion models.
- IM algorithms (Section 4). In order to solve the difficulties of the IM problem proposed in Chapter 3, this chapter elaborates the existing research solutions in more detail. The greedy methods are first reviewed, and then the existing IM algorithms are divided into four categories: simulation-based, heuristic based, sampling-based, and meta-heuristic based, and a thorough theoretical comparison is made.
- Context-aware IM algorithms (Section 5). In this section, we review the extension problem for the IM problem, which makes the final seed set obtained more efficient by considering the user's location, interests, topics of information dissemination, dissemination time, and so on. Since there are more classifications in this part, this review focuses on three main categories: location, time, and topic.

A large number of reviews already exist on influence analysis under social networks. Although these reviews [6–8] focus on IM algorithms, they are outdated and lack the most recent IM algorithms. [9] is a comprehensive experimental study of the latest IM algorithms without any theoretical analysis, and [10] disputes this experiment. Although [11] first examines the issue of context-aware IM and offers a detailed theoretical analysis of current IM algorithms, they only use greedy algorithms to classify IM techniques. [12] does not classify based on the greedy algorithm, but a completely new classification of IM methods. [13] focuses on an algorithmic framework for univariate and multivariate networks-based theoretical analysis of IM algorithms. Additionally, a lot of work has gone into comparing characteristics. [14] introduces IM algorithms from the direction of optimization of algorithms to IM algorithms. [15] focuses on the location-driven influence maximization problem and is not a comprehensive introduction to the influence maximization problem. [16] considers the user's behavior or attitude and summarizes the influence maximization method for user behavior perception.

A variant of the classical IM problem, the context-aware IM-based problem (i.e., combining the time limit of propagation, the geographic location limit of the user, the topic of the propagated message, etc.) has made a great contribution to the practical meaning of the IM problem and the effectiveness of the seed nodes. [11] first summarized the context-aware IM-based problem and the related algorithms, introducing a total of six modules: location-aware, time-aware, topic-aware, dynamic-aware, competition, and other factors. This review is also summarized for this aspect, although it is not as comprehensive as [11], focusing only on time, location, and topic, but we have significant differences in the two parts of time and location. First, the IM algorithm part of this part of time perception is analyzed in more detail, while the elaboration of the algorithm part of [11] is basically similar to that of the diffusion model part, focusing only on the discrete-time perception model and the continuous-time perception model, and not specifically on the algorithmic ideas of this part. Second, for the location-aware part, we divide it into two parts: the online location-aware influence maximization (OLIM) problem and the offline billboard

influence maximization (OBIM) problem, for the elaboration of the OBIM problem, which does not exist in [11].

In addition to the above differences, this review has three major contributions that are not found in other reviews, including [11]. The specific three major contributions are as follows:

- In addition to the classical diffusion models, we propose for the first time two major classes of context-aware diffusion models and deep learning-based diffusion models. And the characteristics of the three major classes of diffusion models are compared, such as Table 1 (Classical diffusion models), Table 2 (Context-aware diffusion models) and Table 3 (Deep Learning-based diffusion models).
- The theoretical difficulties of the IM problem are classified in more detail, and the two major male difficulties, Influence Maximization Problem and Propagation Influence Problem, are summarized more clearly and unambiguously. The Influence Maximization Problem is to find the smallest seed set that maximizes the social impact. Propagation Influence Problem refers to the process of solving the IM problem, which requires overcoming the problem of estimating the social influence of a given seed set.
- For the classification of classical IM algorithms, in addition to the classical three major classifications: simulation-based, heuristic-based, and sampling-based, this review adds for the first time a powerful tool for solving NP-hard problems, a meta-heuristic-based approach. For context-aware IM algorithms, Offline Billboards IM (OBIM) is added to location-aware IM algorithms for the first time, and Online Location-aware IM (OLIM) is classified like classical IM algorithms. Of course, finally, we perform a theoretical comparison analysis of all IM algorithms, as shown in Table 4, Table 5 and Table 6.

## 2. The influence maximization problem

### 2.1. Problem definition

**Definition 1** (*Influence Maximization (IM) [5]*)**.** The influence maximization problem is to find the set of influential users (seed set) $S \subseteq V$ consisting of $k < |V|$ nodes, given a social network $G = (V, E)$ and a positive integer $k$, that can maximize the number of influenced users in the social network $G$ through the propagation of the diffusion model. Where $V$ is the set of nodes in $G$ (i.e., users) and $E$ is the set of (directed/undirected) edges in $G$ (i.e., social connections between users). The problem can be described by the following equation:

$$IM_M(G, k) = \text{argmax}_{e \subseteq V, |e|=k}\, \sigma_M(e, G) \tag{1}$$

where $\sigma$ is a function that calculates the spread of influence for a given set of nodes, and it represents the spread of influence achieved by activating the set of nodes in e.

### 2.2. Problem hardness

From the problem definition, it is clear that solving the IM problem first requires estimating the propagation impact $\sigma(S)$ of a given seed set $S$, and then making this propagation impact maximized. But both of these steps are theoretically difficult. We refer to the theoretical difficulty of estimating the propagation impact $\sigma(S)$ of a given seed set $S$ as the propagation impact problem, and the theoretical difficulty of maximizing the propagation impact as the IM problem.

- **Propagation Influence Problem**. Regarding the theoretical difficulty of estimating the spreading influence $\sigma(S)$ for a given seed set $S$, [17] have shown that the problem is #p-hard.

**Theorem 1.** *Under the Independent cascade model (ICM), the estimated propagation influence $\sigma(S)$ for a given set $S$ of already activated users is #P-hard [17].*

**Theorem 2.** *Under the Linear threshold model (LTM), the estimated propagation influence $\sigma(S)$ for a given set $S$ of activated users is #P-hard [18].*

Given the above theorem, we cannot find the optimal solution in polynomial time unless N = NP, so is much research dedicated to overcoming this theoretical difficulty. For example, simulation-based, heuristic-based, sampling-based, etc. The details can be found in Chapter 4, Section 2, Classification of existing IM algorithms.

- **Influence Maximization Problem**. Kempe et al. [5] have demonstrated the theoretical difficulties of maximizing the propagation impact $\sigma(S)$.

**Theorem 3.** *The IM problem is NP-hard under the Independent cascade model (ICM), Linear threshold model (LTM) and Triggering Model (TRM) [5].*

## 3. Taxonomy of diffusion models

As an important part of the influence maximization problem, the diffusion model is to solve the problem of how to spread information between nodes in a social network. In the diffusion model, at any time stamp $t$, any node $u \in V$ has two states, active and inactive. In the original state $t = 0$, except for the seed set $S \subseteq V$, other nodes in $V$ are in an inactive state. The subsequent process starts from the seed node. Each diffusion model deactivates the neighbor nodes of the seed node according to its own rules and the behavior of neighbor nodes. These activated nodes continue to affect their neighbors until no new nodes can be used. Activation ends the propagation process. The research on diffusion model is divided into the following three categories: (1) Classical Models; (2) Context-aware Models; (3) Deep learning-based diffusion Models.

### 3.1. Classical models

Classical diffusion models are usually divided into threshold models, cascade models, trigger models, and epidemic models. These diffusion models are stochastic models. They flow the propagated information from one node to its neighbors at each point in time according to some probabilistic rules [5]. Table 1 summarizes the analysis in this section, and the details are explained in the following subsections.

#### 3.1.1. Threshold model (TM)

Threshold models were first proposed by [19]. Any threshold model has a threshold or a set of thresholds that are used to distinguish a range of values for the behavior predicted by the model. One of the most frequently used models among the threshold models is the Linear threshold model (LTM). In LTM, each edge $e = (u, v) \in E$ has a weight $W_{u,v}$, which is the important percentage of node $u$ inside all incoming neighbor nodes of node $v$. Each node $v$ also has an influence threshold $T_v$, which follows a uniform distribution of $[0, 1]$ and does not change once determined. At $t = 0$, the seed node set is activated first as the initial node and all other nodes are inactive. At a certain moment in $t \geq 1$, each inactive node $v$ needs to add all the active in-neighbors to its linear weighted sum, and then check whether it has reached its affected threshold $T_v$, which is $\sum_{v \in E(u,v)} w_{u,v} \geq T_v$. If so, node $v$ is activated at time $t$. At a certain moment, no more new nodes are activated, then the propagation process ends.

Depending on the threshold value, other threshold models can be classified into the following categories: (1) majority TM [4,20], where $\left(T_v = \frac{1}{2} D(v)\right)$; (2) small TM [21], where the threshold value $T_y$ is a very small constant; (3) Separated TM [22,23], with the exception of maintaining the threshold and diffusion process independently for each approach, this model is comparable to the LTM model; (4) unanimous TM [4], where $\left(T_v = D(v)\right)$, $D(v)$ in the above equation denotes the degree centrality of node $v$. [20] proposed LT-C (linear threshold with color), which focuses on more than just the impact between users, but the adoption rate of the product.

**Table 1**
Comparison of the properties of classical diffusion models.

| Classical diffusion model | Categories | Social network | | Submodularity | | Activation | |
|---|---|---|---|---|---|---|---|
| | | Directed | Weighted | Diminishing return | Monotone | Time constraint | Multiple activation |
| Threshold Model (TM) | Linear TM (LTM) | √ | √ | √ | √ | × | × |
| | Majority TM | √ | √ | √ | √ | √ | × |
| | Small TM | √ | √ | √ | √ | × | × |
| | Unanimous TM | √ | √ | × | × | × | × |
| | Separated TM | √ | √ | × | × | × | × |
| | LT-C | √ | √ | √ | √ | × | × |
| Cascading Model (CM) | Independent CM (ICM) | √ | √ | √ | √ | × | × |
| | Weight CM | √ | √ | √ | √ | × | × |
| | DICM | √ | × | √ | × | √ | × |
| | Opinion CM | √ | √ | √ | × | √ | √ |
| | ICM-NO | √ | √ | √ | √ | × | × |
| Triggering Model (TRM) | Decreasing CM | √ | √ | √ | √ | × | × |
| Epidemic Model (EM) | SIR | √ | √ | × | √ | × | × |
| | SIS | √ | √ | × | √ | × | √ |
| | SIRS | √ | √ | × | √ | × | √ |

#### 3.1.2. Cascading model (CM)

Goldenberg et al. [24,25] first introduced Cascade models to the field of viral marketing. the Independent cascade model (ICM) is the most widely used model within the Cascade model. Each edge $e = (u, v) \in E$ of this model has an activation probability $p_{u,v} \in [0, 1]$, and the activation probability formula is $p_{u,v} = 1 - (1 - p)^{w_{uv}}$, where $w_{uv}$ is the weight of the edge [26].

At $t = 0$ moment, the seed node set is activated first as the initial node and all other nodes are inactive. At some point in $t \geq 1$, an inactive node $v$ is activated once and only once at $t$ by all incoming neighbors activated at the previous moment with $p_{u,v}$ activation probability, i.e., each incoming neighbor node of node $v$ that was activated at the previous moment can only activate it independently, and these incoming neighbors will remain active and stop activating afterwards. When a node fails in its activation attempt and other incoming neighbor nodes of node $v$ do not successfully activate node $v$ at time $t$, node $v$ remains inactive at time $t$. The propagation process ends only if, at some point, no more new nodes are activated.

The WCM (weighted cascade model) [27] is an extension of the traditional IC model, which also follows the above rules but the concept of node attributes is added. That is, in addition to an activation probability $p_{u,v}$ for each edge, each node has a weight with a non-negative value $w_u$, which is calculated independently of the network structure based on its attributes. [28] proposed a Each node $u$ in the Dynamic Independent Cascade Model (DICM) has a random variable $Xu$ that follows a Bernoulli distribution. When $X_u = 1$, it means that node $u$ has been successfully activated. Besides, there are some ICM-based diffusion models, such as ICM with negative opinions (ICM-NO) [29], Opinion CM with positive and negative opinions [30].

#### 3.1.3. Triggering model (TRM)

Additionally, [5] suggested Trigger models, emphasizing that TM and CM are TR's exceptions. Each node $v$ in the Trigger model has a threshold $T_v$ and a distribution function $f_v$. The probability that this distribution function transfers to a subset of its neighbors indicates how much the subset of neighbors can influence node $v$. Based on the distribution of the subset of neighbors of node $v$, the TR model chooses a random "trigger set" for each example of the diffusion process. At the moment of $t = 0$, the seed node set is activated first as the initial node, and the other nodes are in an inactive state. At a certain time of $t \geq 1$, if the inactive node $v$ has an activated neighbor node at time $t - 1$ in the selection trigger set, then at time $t$ node $v$ will be activated. At a certain moment, no more new nodes are activated, then the propagation process ends.

In place of the trigger model, [31] put out the Decreasing Cascade (DC) model, which is a more comprehensive model. The influence probability of node $u$ on node $v$ is now defined by this model as $p_v(u, S_u)$, where $S_u$ represents node $u$'s subset of active neighbors. To capture diminishing returns, DC enforces $p_v(w, S) \geq p_v(w, T)$, for $S \subseteq T$.

#### 3.1.4. Epidemic model (EM)

Epidemic transmission has a close similarity to information dissemination in social networks. In this model, information spread in social networks can be considered as a disease in the epidemic transmission process [32]. In accordance with the characteristics of the model cascade, three epidemic models have been proposed: susceptible to infectious disease recovery (SIR), susceptible to infectious disease susceptible (SIS), and susceptible to infectious disease recovery susceptible (SIRS) [23]. The susceptible–infected (SI) model is the simplest. It says that each node can be in one of two states: susceptible or infected. When a node is in the vulnerable state, it can be infected. Once a node $u$ is infected, it will always be infected and spread on its own, since it only has one chance to affect a neighboring node $v$ that is in the susceptible state. susceptible–infected–susceptible (SIS) model [33] is similar to SI, except that it considers that an infected node $u$ can return to the susceptible state with The susceptible infected recovered (SIR) model [23] is roughly the same as SI, but it assumes that nodes have three states: susceptible, infected, and recovered. It assumes that an infected node $u$ can restore health and develop immunity to disease with a certain probability, which means that node $u$ will never be infected again, and the rest is the same as SI. recovered susceptible (SIRS) extends the SIR model by assuming that an infected node $u$ will recover health with a certain probability and then may become susceptible again.

### 3.2. Context-aware models

The classical diffusion model only considers the activation probability as a factor in formulating the influence propagation rules. However, the activation between nodes does not simply depend on the activation probability between two nodes, and the seed set found in this way does not have a certain validity. Therefore, it is necessary to consider the user's geographical location, the time of information dissemination, the topic of information dissemination, the user's interest, the group effect, product competition, and other factors in the dissemination process. For different context-aware IM problems, the corresponding diffusion models are different. In this paper, we focus on three major context-aware IM problems (see Section 5), which are location, time, and topic. For the topic part, the classical diffusion model is mainly extended, and for the location and time parts, the classical diffusion model is

**Table 2**
Comparison of the properties of context-aware diffusion models.

| Context-aware model | Categories | Diffusion model | Social network | | Submodularity | | Activation | |
|---|---|---|---|---|---|---|---|---|
| | | | Directed | Weighted | Diminishing return | Monotone | Time constraint | Multiple activation |
| Location-aware (LAM) | Online location-aware | ICM | √ | √ | √ | √ | × | × |
| | | LTM | √ | √ | √ | √ | × | × |
| | Offline Billboards | OI | √ | – | × | × | × | × |
| | | MI | √ | – | – | – | × | × |
| | | LI | √ | – | – | – | × | × |
| Time-aware (TAM) | Discrete-time | IC-M | √ | √ | √ | √ | √ | × |
| | | CTIC | √ | √ | – | – | √ | × |
| | | LAIC | √ | √ | √ | √ | √ | × |
| | Continuous-time | TV-IC | √ | √ | √ | √ | √ | × |
| | | TV-LT | √ | √ | √ | √ | √ | × |
| | | DynaDiffuse | √ | √ | √ | √ | √ | × |

improved or a new diffusion model is proposed based on the classical diffusion model. So this section mainly introduces the Location-aware Model (LAM) and Time-aware Model (TAM). Table 2 summarizes the analysis in this section, and the details are analyzed in the following subsections.

### 3.2.1. Location-aware model (LAM)

The location-aware context-based problems are divided into two categories: online location-aware impact maximization (OLIM) and offline billboard impact maximization (OBIM). Among them, the OLIM problems use the classical diffusion models, and the most frequent diffusion models are the Independent Cascade Model (ICM), the Linear Threshold Model (LTM), and the Maximum Impact Tree Model (MIA) [18] respectively. New diffusion models are proposed for the OBIM problem, which are the One-time impression (OI) model [34], Multi-time impression (MI) model [35] and Logistic influence (LI) model [36].

- **One-time impression (OI) model**. [34] presented the OI models, which specify that given a trajectory database $\mathcal{T}$ consisting of a set of trajectories $t = \{p_1, p_2, \ldots, p_{|t|}\}$, a billboard $b$ with location information b.loc and a given threshold $lambda$. For each trajectory $t$, if $\exists p_i \in t$ such that Distance $(p_i,$ b.loc $) \leq \lambda$, i.e., the distance from $p_i$ to b.loc is less than or equal to the threshold $\lambda$, then it can be said that this trajectory $t$ is influenced by the billboard $b$.
- **Multi-time impression (MI) Model**. [35] proposed MI models which specify that given a trajectory database $\mathcal{T}$ consisting of a set of trajectories $t = \{p_1, p_2, \ldots, p_{|t|}\}$, the set of billboards $B = \{b_1, b_2, \ldots, b_l\}$, size $(b_i)$ is the panel size of billboard $b_i$. If the billboard $b_i \in B$ can affect the trajectory $t$, then set pr $(b_i, t)$ to a uniform value, where pr $(b_i, t) =$ size $(b_i)/A$, $A = \max_{b_i \in B}$ size $(b_i)$. Finally, [35] give the formula for the effect of the billboard set $S \subseteq B$ on the trajectory $t$: $\mathrm{pr}(S, t) = 1 - \bigsqcup_{b_i \in S}(1 - \mathrm{pr}(b_i, t))$.
- **Logistic influence (LI) model**. [36] proposed LI models, which specify that given a trajectory database $\mathcal{T}$ consisting of a set of trajectories $t = \{p_1, p_2, \ldots, p_{|t|}\}$, a set of billboards $B = \{b_1, b_2, \ldots, b_l\}$ and the Bernoulli random variable $I(b_i, t)$, which indicates whether the billboard $b$ affects the state of the trajectory $t$. When $I(b_i, t) = 1$, it indicates a high influence. The logistic influence model evaluates the effective influence P (S, t) of a set of billboards $S \subseteq B$ on the trajectory $t$ by using equations based on the logistic function.

### 3.2.2. Time-aware model (TAM)

The classical diffusion model is time-independent, which means that since there are no new nodes activated in the traditional diffusion model, the diffusion process comes to an end. However, the diffusion process is often time-limited, so it is necessary to consider the time factor. There are approximately two categories of current research: (1) Discrete-time Model; (2) Continuous-time Model.

- **Discrete-time Model**. Discrete-time models are essentially similar to IC and LT because they are still in discrete time steps. Therefore, this type of model mainly models the diffusion process between nodes as discrete random variables at different time steps to achieve the purpose of extending the IC model.
  Based on the traditional ICM model, the IC-M model [37] was introduced. In the IC-M model, node $x$ has one chance to deactivate node $y$ with probability $p_{x,y}$, and node y has many chances to contact node $x$ with probability $m(x, y)$. Under this model, the seed set S is selected so that the expected influence of the seed set within $\tau$time steps of the random process is the largest.
  [38] proposed Continuously activated and Time-restricted IC (CT-IC) by adding two properties of time constraint and continuous activation to the traditional ICM model. The difference between this model and the IC model is that: (1) each active node can repeatedly activate its neighbors; and (2) the activation process will not stop until a given time $T$. The CT-IC model satisfies the monotonicity and sub-modularity of the influence propagation, so it can be guaranteed that the greedy algorithm under the CT-IC model has an approximate ratio of $(1 - 1/e)$. First, each seed node is activated at moment 0 and activates neighboring nodes at moments $t = 1, 2, 3, \ldots$. $A_t$ is the set of nodes activated at time $t$, where $A_0 = S$. Each node $u \in A_t$ that is active activates its neighbor nodes $u$ with probability $pp_{t-t_u}(u, v)$, where $t_u$ is the time to activate the neighbor nodes $u$. where the activation probability is defined as follows: $pp_t(u, v) = pp_0(u, v) \cdot \exp(-\alpha_u x)$, where $\alpha_u$ denotes the rate at which the impact of node u on its neighbors decreases.
  with the same concept as [38–40] suggested a delay independent cascade (LAIC) model that incorporates information about influence propagation delay into the IC model. In this model, when node $u$ is activated for the first time at step $t$, it activates the currently inactive neighbor node $v$ at step $t + \delta_t$ with probability $\mathcal{P}_{uv}\mathcal{P}_u^{lat}(\delta_t)$, where $delta_t$ is the influence delay. Additionally, a node can only be activated once. A node will be triggered at the earliest activation time if it is impacted by more than one neighbor; the other activations will be disregarded.
- **Continuous-time Model**. The continuous time model is more consistent with real-world scenarios because the diffusion process between nodes is inherently continuous in time. Given a predefined stopping time $T > 0$, if no more nodes are activated before the stopping time $T$, then the diffusion process is stopped. And given the activation time $t_u$ of node $u$, the probability that node $u$ activates node $v$ at any time period $t_v > t_u$ is $p(t_v \mid t_u; \alpha_{u,v})$, where $\alpha_{u,v}$ is used to determine the influence of node $u$ to node $v$ and the influence strength of node $v$.
  [41] first proposed a continuous-time independent cascade model, which models the diffusion process as a discrete network of continuous time processes occurring at different rates. They modeled diffusion using only time-dependent propagation probabilities

**Table 3**
Comparison of the properties of deep Learning-based diffusion models.

| Algorithm | Year | Contribution | Comparable algorithm |
|---|---|---|---|
| DeepCas [44] | 2017 | Using an end-to-end deep learning approach to predict the future size of cascades | feature based node embedding graph kernel |
| IEDP [45] | 2017 | The information embedding-based model IEDP is proposed to predict the information diffusion process | ICM GDM model CSDK |
| Topo-LSTM [46] | 2017 | Representation learning methods are proposed for exploring cascade structures and thus for diffusion prediction | EmbeddedIC DeepCas DeepWalk IC-SB |
| Embedded-IC [47] | 2016 | The embedded cascade model Embedded-IC is proposed to embed users of social networks into the potential space to extract more robust diffusion probabilities than those defined by classical graph learning methods | ICM Netrate CTIC |

between pairs of nodes, propagation rates, and infection times, without using a priori probabilities of infection that depend on unknown external factors.

[42] introduced the Instantaneous Variable Independent Cascade (TV-IC), a more broad time-aware diffusion model based on ICM. The model assumes that node $x$ is active at time step $t1$ and communicates with node $y$ through edge $(x, y)$. At time step $t2$, the propagating influence reaches node y, but node $y$ is not activated at this time. Whether node $y$ is activated at the $t2$th time step depends on the time difference $(t2, t1)$. [42] likewise proposed the first time-aware model based on the LTM called time-varying linear threshold (TV-LT). Because of the submodularity of the diffusion functions of both models, the algorithm can provide a provable approximation guarantee.

[43] proposed a continuous-time dynamic diffusion model (DynaDiffuse) for dynamic diffusion networks, able to consider the influence of a set of nodes when estimating their spread of influence over a given time horizon. They assumed that in this model, once a node is affected, this state continues until the end. The process ends when there are no remaining unaffected nodes or when a given time $T$ has elapsed.

### 3.3. Deep learning-based diffusion models

Earlier work assumed that diffusion models were given, such as the Independent Cascades Model (ICM) and Linear Thresholds Model (LTM). Some research attempt to develop a diffusion model from existing cascade data by constructing valuable variables, such as network structure and temporal information [17], social roles of user nodes [48], and user interaction of nodes [49], to forecast node activation probability. Even while machine learning techniques clearly improve diffusion prediction accuracy, creating features still needs a lot of manual labor and specific domain expertise. With the rapid development in the field of neural networks, recent works [44–47]has started to use deep learning for diffusion modeling, thus avoiding the heavy feature engineering. Table 3 summarizes the analysis in this section, and the details are analyzed in the following subsections.

Li et al. [44] proposed the end-to-end deep learning method DeepCas to predict the future size of the information cascade. An induced subgraph on the active nodes during each time interval is used to model the cascade initially. The subgraph is then divided up into a few random walk pathways, and its embedding vector is learned using a gated recursive unit (GRU). The magnitude of the upcoming cascade is thus projected using the embedding vector of this subgraph. Alternative node embedding and graph embedding techniques, as well as feature-based machine learning techniques, are all outperformed by the end-to-end predictor DeepCas.

Gao et al. [45] suggested employing an information embedding-based propagation prediction (IEDP) model rather than an explicit graph-based method to the extracted social structure to understand the temporal dynamics of diffusion from observations. They turned the challenge of temporal propagation prediction into a goal of learning spatial probabilities in the embedding space, and by embedding the diffusion content and the relative positions of users in the possible embedding space, they calculated the probability of diffusion propagation in IEDP. The estimation of information diffusion in the prospective embedding space is also suggested, along with an effective margin-based optimization procedure.

[46] proposed Embedded-IC, an embedded version of the independent cascade model, to predict information dissemination in online social networks. They classified nodes so that in a diffusion cascade, active nodes can communicate information to dormant nodes. It acquires a vector for each role to serve as the node's embedding. It then built an activation model based on the proximity between the receiver embedding vector of the inactive node and the sender embedding vector of the active node. But regardless of the network structure, the activation time for enabling the timestamp t between inactive nodes and all existing active nodes is t. As a result, each sender's (or active node's) embedding is learned independently of the diffusion topology.

[47] proposed a representation learning method for predicting information diffusion and designing the Topological Long Short-Term Memory (Topo-LSTM) method. Designed specifically for the job of predicting node activation, Topo-LSTM is a Dynamic Directed Acyclic Graph (DAG) structure for processing diffusion topologies. A dynamic DAG serves as its input, and its output is a topology-aware embedding created by each DAG node.

## 4. Overview of IM algorithms

### 4.1. The greedy approach

Although the IM problem is NP-hard, it is not feasible to find an optimal solution to the problem. However, if the expected impact diffusion function $\sigma(S)$ is submodular, then the problem can be solved

by solving an approximate solution that approximates the optimal solution. Solving for the approximate solution gives rise to the concept of an approximate ratio, which is the ratio of the expected impact diffusion to the actual impact diffusion. A function is said to be submodular if it satisfies both the monotone increasing and diminishing return properties.

**Property 1** (*Monotone Increasing*). *When the objective function $\sigma(S)$ satisfies $\sigma(S) \leqslant \sigma(T), S \subset T$, then this function is monotonically increasing.*

**Property 2** (*Diminishing Return*). *When the objective function $\sigma(S)$ satisfies $\sigma(S \cup u) - \sigma(S) \geqslant \sigma(T \cup u) - \sigma(T), \forall u \in T$ and $S \subset T$, then this function is of decreasing returns.*

In more general terms, monotone increasing implies that the overall propagation impact does not decrease as nodes are added to the seed set $S$. According to the theory of diminishing returns, the marginal returns of node $u$ and a portion of the seed set are consistently higher than or equal to the marginal returns of the seed set. And Kempe et al. [5] have shown that under the classical diffusion model, the expected impact diffusion function $\sigma(S)$ is submodular. Therefore, under the classical diffusion model, it is possible to solve the IM problem by finding approximate solutions that approximate the optimal solution.

**Theorem 4.** *Under the Independent cascade model (ICM), Linear threshold model (LTM) and Triggering Model (TRM), the expected influence diffusion function $\sigma(S)$ is submodular [18].*

With the above entry points, [5] proposed a greedy algorithm 1, which is the first solution to the IM problem, to find the $k$ irrelevant seed nodes that make its propagation impact maximum. The algorithm first detects that the seed set is empty, iterates over $k$ steps to estimate each node's marginal returns, then adds the node $x$ with the biggest increase in marginal effect to the seed set $S$ at the end of each iteration [50]. The greedy algorithm ensures that the expected influence between the optimal seed set $S$ and the actual influence of the greedy seed set $S^*$ satisfies $\sigma(S) \geqslant \left(1 - \left(1 - \frac{1}{k}\right)^k\right) \sigma(S^*)$, i.e., the approximation ratio is $\left(1 - \left(1 - \frac{1}{k}\right)^k\right)$, but [51] pointed out that the approximation ratio for the usual case would be denoted by $\left(1 - \frac{1}{e}\right)$. Because when $\lim_{k\ rightarrow\infty} \left(1 - \left(1 - \frac{1}{k}\right)^k\right)$ and $k > 0$, $\left(1 - \frac{1}{e}\right) < \left(1 - \left(1 - \frac{1}{k}\right)^k\right)$.

**Algorithm 1:** Greedy Algorithm [5]

**Input:** Social graph: $G$; Seed size: $k$;
Influence Function: $\sigma(S)$.
**Output:** Seec set: $S$.

```
S ← ϕ;
for i = 1, 2, ..., k do
    u* ← argmax_{u∈V\S} (σ(S ∪ {u}) − σ(S))
    S ← S ∪ {u*}
return Seed Set S
```

But the greedy algorithm finds the node that gains the most influence at each phase. And you must know the influence diffusion generated by adding each candidate node to the current seed set. But computing $\sigma(S)$ for a seed set S is #p-hard (Section 4. Theorem 1). At that time, no precise algorithm was found to solve this problem, but Monte Carlo simulations of the impact cascade model were performed enough times, that is, a simulation-based method, to obtain an accurate estimate of the impact diffusion. So, on modern servers, finding even a small seed set in a moderately sized network can take days to complete, which is time-consuming.

### 4.2. Classification of existing IM algorithms

We have shown in Section 1 of Chapter 3 that estimating the propagation impact $\sigma(S)$ for a given seed set $S$ is #p-hard, so this section will provide a fine-grained classification for studies that overcome this theoretical difficulty. They are classified into the following four categories according to the direction they take to overcome this theoretical difficulty: (1) Simulation-based Approaches; (2) Heuristic-based Approaches; (3) Sampling-based Approaches; and (4) Meta-heuristic-based Approaches. Tables 4 and 5 summarize the analysis in this section, and the details are analyzed in the following subsections.

#### 4.2.1. Simulation-based approaches

The core idea [5] of the simulation-based method is to use Monte Carlo (MC) simulation to estimate the spread influence of the user set $\sigma(S \cup \{X\})$ after each addition of node $x$, the final output is the number of active users $N(S \cup \{X\})$. And [7] by analyzing the relationship between r and relative error, it was pointed out that only when r is large enough, the error will be small enough. Therefore, the existing optimization techniques for simulation-based methods are mainly divided into two categories: (1) Decreasing the number of MC simulation; and (2) Reducing the search space for MC simulations. The two distinct optimization strategies will be discussed in the following paragraphs.

- **Decreasing the number of MC simulations**. [52] proposed the Cost-Effective Lazy Forward (CELF) approach uses the submodule property of the impact maximization objective to greatly reduce the number of evaluations of the impact propagation of vertices. The core idea of the algorithm is that the marginal gain of a node in the current iteration cannot be greater than its marginal gain in the previous iterations. CELF maintains a table $\langle u, \Delta_u(S)\rangle$ in decreasing order of $\Delta_u(S)$, where S is the current seed set and $\Delta_u(S)$ is the marginal benefit. $\Delta_u(S)$ will only be re-evaluated against the top node when needed. For the re-evaluated table, if a node is still at the top, it is the next seed node. Their experimental results show that CELF optimization can achieve a speedup of up to 700 times in selecting seed vertices. [53] proposed the CELF++ method based on CELF, which improves the efficiency by 35%–55% compared to CELF. The idea is that if a node $x$ was selected as a seed node in the previous round, then this node does not need to be considered when calculating its marginal benefit in the next iteration. [54] proposed UBLF based on CELF++, it computes an upper bound on the expected influence of node $u$ by multiplying the propagation probability matrix (which is also a sparse matrix) several times, thus avoiding most of the initial iterations of CELF++. And through the design of the propagation probability matrix, the method can be extended to more general diffusion models.
- **Reducing the search space for MC simulations**. Most of the models devoted to this part of the study are by dividing the network and thus reducing the search space of MC simulations. [55] proposed the community-based greedy algorithm (CGA), which uses a divide-and-conquer approach. First, the social network is divided into several sub-networks, and then influential users are searched within the sub-networks separately. However, CGA needs to model the marginal benefits of each node in the community and is not applicable to large-scale graphs. The idea of community-based influence maximization (CIM) proposed in [56], like CGA, is to reduce the search space by dividing the network. [60] proposed a community-based context-aware influence maximization algorithm (C2IM) in combination with context-awareness. Its contribution is mainly twofold: (1)using the community framework to reduce the search space and improve the time efficiency, and (2) by considering the user's interest to enhance the effectiveness of seed nodes. C2IM first divides the

**Table 4**
A comparison of the existing influence maximization algorithms under classical diffusion models - I.

| Approaches | Categories | Algorithm | Year | Diffusion model | Time complexity | Approximation | Comparable algorithm |
|---|---|---|---|---|---|---|---|
| Simulation-based | Minimization of number of MC simulations | Greedy [5] | 2003 | LTM ICM TR | $O(kNMI)$ | $1-1/e-\epsilon$ | High Degree & Random |
| | | CELF [52] | 2007 | LTM ICM TR | $O(kNMI)$ | $1-1/e-\epsilon$ | General Greedy |
| | | CELF++ [53] | 2011 | LTM ICM TR | $O(kNMI)$ | $1-1/e-\epsilon$ | CELF |
| | | UBIF [54] | 2015 | LTM ICM | $O(kINM)$ | $1-1/e-\epsilon(I)$ | CELF & High Degree, DegreeDiscount, MIA & SP1M |
| | Reduce the search space of MC simulations | CGA [55] | 2010 | ICM | $O\left(M+IM_C\left(N(Z-C)+k\left(C+N_C\right)\right)\right)$ | $1-e^{-1/(1+\delta_c)}$ | MixedGreedy, NewGreedy, DD & Random |
| | | CIM [56] | 2014 | HDM | – | N.A | CDH-Kcut, CDH-SHRINK, EGA & DH |
| | | CoFIM [57] | 2017 | WCM | $O\left(k^2\cdot N\cdot k_{max}\right)$ | $1-1/e$ | IMM, TIM+, IPA, & High Degree |
| | | Basic solution [58] | 2017 | LTM | $O\left(3^{n/3}+kn\log_k n\right)$ | N.A | SO,SO+,BF & BF+ |
| | | ComBIM [59] | 2019 | ICM WCM | $\mathcal{O}\left(\ln^2+nd+l^2\right)$ | N.A | Random, DAG1-SPBP, High Degree & BSSA |
| | | C2IM [60] | 2019 | LTM ICM | $O((\|V\|+\|E\|)*\log(\|V\|+\|E\|))$ | N.A | Random, CIM, HighDegree & TIM |
| | | TI-SC [61] | 2020 | ICM | $O(N\log N+M+RN+(k-1)Rc)$ | N.A | LIR, High Degree, k-core, CI & PHG |
| | | CDIM [62] | 2020 | ICM | $O\left(kn^2\cdot\frac{1}{2\epsilon^2}\log\frac{n}{\eta}\right)$ | $1-1/e-\epsilon$ | IM, Heuristic, DIV(LPA), DIV(Louvain) |
| | | CSR [63] | 2022 | ICM SIR | – | N.A | ENC, EVC, GC, GLR, EC & LID |
| Heuristic-based | Influence ranking | Degree discount [64] | 2009 | LTM ICM WCM | $O(k\log N+M)$ | N.A | CELF, Random, High Degree, NewGreedyICM & Distance |
| | | PageRank [65] | 2012 | LTM ICM WCM | – | N.A | General Greedy, MixedGreedy & NewGreedy |
| | | GROUP-PR [66] | 2014 | ICM | $O(kMN)$ | N.A | CELF, IRIE, PMIA, PageRank & High degree |
| | | LIR [67] | 2017 | LTM ICM | $O\left(M+kN'\right)$ | N.A | High Degree, NewGreedy, distance & MixedGreedy |
| | | HybridRank [68] | 2018 | SIR | $O(\|N\|+\|M\|)$ | N.A | Eigenvector, K-shell & High degree |
| | | LGIM [69] | 2019 | ICM | $O(pop*M\log N)$ | N.A | DD, PMIA, IMM & DDSE |
| | | IMMRA [70] | 2020 | ICM | $O(k^*T^*n)$ | N.A | General Greedy |
| | Diffusion model reduction | SPM/SP1M [71] | 2006 | ICM | $O(kNM)$ | $1-1/e$ | High Degree, PageRank, Closeness & Betweenness |
| | | MIA/PIMA [18] | 2010 | ICM | $O\left(Nt_{i\theta}+kn_{0\theta}n_{i\theta}\left(n_{i\theta}+\log N\right)\right)$ | $1-1/e$ | General Greedy, DD, SPIM, PageRank & Random |
| | | LDAG [18] | 2010 | LTM | $O\left(Nt_\theta+kn_\theta m_\theta\left(m_\theta+\log N\right)\right)$ | N.A | General Greedy, SPIN, DD & PageRanK |
| | | SIMPATH [72] | 2011 | LTM | $O\left(kINP_\theta\right)$ | N.A. | High Degree, PageRank, CELF, LDAG & CELF++ |
| | | IPA [73] | 2013 | ICM | $O\left(\frac{NO_{v,n_{iu}}}{c}+k^2\left(\frac{O_{v,n_{iu}}}{c}+(c-1)\right)\right)$ | N.A | CELF, PMIA, SD & Random |
| | | EASYIM [74] | 2016 | LTM ICM WCM | $O(kD(N+M))$ | N.A | CELF++, TIM+, IRIE & SIMPATH |
| | Others | RLIM [75] | 2020 | ICM | – | N.A | PHG, LIR, CI & k-core |
| | | CTT [76] | 2021 | ICM | – | N.A | DIM & TESA |

(*continued on next page*)

network into sub-networks using the community detection algorithm CDA, and then designs A non-ideal node finder (NDF) is designed to filter out non-ideal nodes. Finally, the seed selection algorithm (SSA) is used to select the most influential C2IM performs better than CIM in influence diffusion and faster than TIM.

For optimizing influence in networks, a community-based approach (CoFIM) [57] has been proposed. They separated the two stages of influence propagation into seed expansion and intra-community propagation. In other words, the seed nodes are initially permitted to spread throughout several communities before spreading within communities that are separate from one

**Table 4** (*continued*).

| Approaches | Categories | Algorithm | Year | Diffusion model | Time complexity | Approximation | Comparable algorithm |
|---|---|---|---|---|---|---|---|
| Sampling-based | Forward influence | NewGreedy [64] | 2009 | ICM | $O(kIM)$ | $1-1/e-\epsilon(r)$ | CELF, Random, SD & HighDegree |
| | | StaticGreedy [77] | 2013 | ICM | $O\left(\frac{KMN^2\log_N K}{\epsilon^2}\right)$ | $1-1/e-\epsilon$ | CELF, PMIA, SP1M, High Degree & DD |
| | | PRUNEDMC [78] | 2014 | ICM | $O\left(\frac{KMN^2\log_N K}{\epsilon^2}\right)$ | $1-1/e-\epsilon$ | IRIE, PMIA, Degree |
| | | SKIM [79] | 2014 | ICM | $O\left(\frac{KN^2M\log_N K}{\epsilon^2}\right)$ | $1-1/e-\epsilon$ | TIM |
| | | TIM/TIM+ [80] | 2014 | LTM ICM TR | $O\left(\frac{k(M+N)\log N}{\epsilon^2}\right)$ | $1-1/e-\epsilon$ | CELF++, IRIE & SIMPATH |
| | Reverse reachable | RIS [81] | 2014 | LTM ICM TR | $O\left(\frac{k(N+M)\log^2 N}{\epsilon^2}\right)$ | $1-1/e-\epsilon$ | – |
| | | IMM [82] | 2015 | LTM ICM TR | $O\left(\frac{(k+l)(N+M)\log N}{\epsilon^2}\right)$ | $1-1/e-\epsilon$ | TIM,TIM+ & IRIE |
| | | SSA/D-SSA [83] | 2016 | LTM ICM TR | – | N.A | IMM & TIM+ |
| | | BKRIS [84] | 2017 | LTM ICM | $O\left(\frac{NM(\log N+\log_N K)}{\epsilon^2}\right)$ | $1-\frac{1}{e}-\epsilon-\epsilon I$ | RIS |
| | | HIS [85] | 2019 | LTM ICM | $O\left(\frac{1}{\epsilon^2}\log\left(\frac{1}{\delta}\right)\frac{n}{\rho^{(h)}(S)}\right)$ | N.A | RIS & SKIS |
| | | RMG [86] | 2020 | ICM | $O\left(k^*(k^*+l+1)(m+n)n^4q^8p_{\max}\cdot\log(nq)/\left(p_{\min}\cdot\epsilon^2\right)\right)$ | $1-1/e-\epsilon$ | Random, Greedy & PMCE |
| | | SRIS [87] | 2020 | ICM | – | $1-1/e-\epsilon$ | CELF & HighDegree |
| | | D-RIS [88] | 2021 | ICM | $O(\theta\times\mathrm{EPV})$ | $1-1/e-\epsilon$ | CELF, RIS, HighDegree, LIR |

**Table 5**
A comparison of the existing influence maximization algorithms under classical diffusion models - II.

| Approaches | Algorithm | Year | Diffusion model | Time complexity | Comparable algorithm |
|---|---|---|---|---|---|
| Meta-heuristic-based | SA [89] | 2011 | ICM | $O(TIM)$ | NewGreedy, DD SAED & Random |
| | SAEDA [89] | 2011 | ICM | $O(Tk\bar{D})$ | NewGreedy, DD Random, SA, MSA, SASH |
| | GNA [90] | 2015 | ICM | – | NewGreedy & SA |
| | DPSO [91] | 2016 | ICM WCM | $O\left(k^2\cdot\log k\cdot n\cdot\bar{D}^2\cdot g_{\max}\right)$ | High Degree, CELF++, SAEDV & PagrRank |
| | DDSE [92] | 2018 | ICM | $O\left(n\cdot k\cdot\bar{D}\cdot g_{\max}\right)$ | CELF, High Degree, & PageRank |
| | DBA [93] | 2018 | ICM | $O\left(n\cdot k\cdot(\log k+\bar{D})\cdot g_{\max}\right)$ | CELF, DDSE & DPSO, SSA |
| | ADPSO [94] | 2019 | ICM | $O\left(k\bar{D}(nC+k\bar{D})g_{\max}+(M+N)\right)$ | CELF, IMPC, SSA & DDSE |
| | GWIM [95] | 2020 | ICM | $O\left(N+g_{\max}\cdot n\cdot V'+g_{\max}\cdot n\cdot k\cdot\langle\bar{D}\rangle\cdot\langle\bar{D}^2\rangle\right)$ | High Degree, CELF++, PageRank, K-shell, DD, DDSE & SAEDV |
| | ASA [96] | 2021 | ICM | – | CELF, DDSE, SAEDV & PageRank |

another. The ComBIM algorithm is proposed for solving the budget influence maximization problem by considering the budget of propagation. [59] introduced the ComBIM method for tackling the budget impact maximization problem by taking propagation budget into account. ComBIM first detects communities to understand the intrinsic structure of the social network, then distributes the entire budget among communities according to the number of nodes and their selection cost, then selects the seed nodes, and finally performs budget transfer by transferring unused budget from one community to the other.

[61] proposed TI-SC, a method for choosing influential nodes that updates scores after each seed node is chosen to minimize overlapping choices. The scores are based on the scoring ability of core nodes with other nodes. [63] proposed a community-based communicator ranking (CSR) and proposed three features to identify bridge nodes as influential: community diversity, community modularity, and community density. They first utilized Louvain's approach to locate the communities in the graph, classified each tiny community into a node, and then used the idea of community variety to locate bridge nodes that connect various communities in the network.

Since existing studies ignore the community diversity of activated nodes, [62] firstly proposed the Community Diversity Impact Maximization (CDIM) problem of finding seed nodes in multiple communities. First, they proposed a metric to measure the impact of community diversity, and second developed two algorithms to solve the CDIM problem, the greedy algorithm and the ceiling-based algorithm. The greedy algorithm guarantees a reasonable approximate solution, and the upper bound-based algorithm minimizes the search space for exploring seed candidates. Finally for estimating the impact of candidate nodes in social networks, they designed the community-aware partial shortest path tree (CPSP). The work of [58] presents the first search problem for the most influential communities in large social networks, aiming to find the highest externally influential communities. They first propose a new community model with desirable properties, i.e., sociality, cohesion, connectivity, and maximization, i.e., maximum kr-Clique communities. Then, a new tree-based index structure, C-tree, is designed to maintain the r-cliques computed offline. finally, four efficient C-tree-based search algorithms, namely, order-based search (SO), improved order-based (SO+) search, best-first (BF)-based search, and best-first (BF+)-based fast search, are designed to improve the search performance of the most influential communities.

- **Summary**. Despite the fact that these optimization strategies aim to improve the performance of greedy algorithms, the significant computational overhead imposed by vast social networks with billions of edges must be taken into account. Existing experiments support experiments such as those in [9,80]. The MC simulation approach is similar to a black box with model generality, so the simulation-based approach is applicable to all diffusion models. However, it is not possible to directly use the properties of

diffusion models to analyze and optimize the impact assessment process and further improve the performance.

#### 4.2.2. Heuristic-based approaches

The fundamental principle behind heuristic-based techniques is that impact approximation scores, as opposed to repeatedly conducting MC simulations, are used to estimate the influence spread $\sigma(S)$ of the seed set S. Unlike simulation-based methods, most heuristics are able to speed up the computational process by exploiting the properties of the corresponding models. But they do not provide any approximate theoretical guarantees. So, simulation-based methods trade time consumption for approximate theoretical guarantees, while heuristic-based methods trade the absence of approximate guarantees for substantial improvements in efficiency. Existing heuristic-based methods can be divided into the following two categories: (1) Node Influence Ranking and (2) Diffusion Model Reduction.

- **Node Influence Ranking**. Because the primary purpose of node influence ranking is to rank the influence of all nodes on the basis of an approximate metrics have been proposed for estimating the diffusion of influence and then to generate the seed set directly from the ranking, selecting a reliable approximate metric is of the utmost importance. In recent years, some simple approximate metrics proposed for estimating influence diffusion: degree, distance centrality [97], and PageRank [65]. But these approximate metrics firstly differ significantly from the actual impact propagation process [64]. Secondly, they did not take into account the influence overlap between various seeds; instead, they just linearly combined the propagation influences of all users in the seed set. Therefore, influence-aware ranking methods have emerged in order to provide a higher quality solution.
  By choosing the node with the highest degree as the seed node each time and reducing the degrees of its neighboring nodes by one at that time, the Degree Discount (DD) [64] was able to overcome the issue of overlapping influence. However, the limitation of this work is that it is based on a uniform IC diffusion model, which is less scalable. [67] proposed an algorithm based on the local index rank (LIR) of each vertex, which improves the Degree Discount (DD) algorithm. Since LIR selects nodes with higher degree than their neighbors, it can ensure that there is no connection between seeds, thus avoiding the "rich-club" effect (the majority of nodes with a high degree are linked to one another). [68] proposed the HybridRank algorithm, which uses a new hybrid centrality metric combined with the topological features of the network to find a set of influential propagators. GROUP-PR [66], like GROUP-PageRank, expanded the concept of PageRank from an individual to a group of users. It also adheres to the greedy framework by first determining the predicted PageRank value of each user and then iteratively locating the seed node with the highest marginal gain, user x, based on the ranking of the values. It avoids the native MC simulation, is more efficient than MC simulation, and is likewise more accurate than PageRank.
  In the actual information propagation process, the intensity of information propagation decreases with time. So [70] first established a three-hop information decay propagation model by capturing the factors affecting decay. The three-hop technique's effectiveness is then predicted, and an impact maximization algorithm (IMMRA) based on the new model and the three-hop approach is then suggested. The technique enhances the quality of the seed set as well as time efficiency. The LGIM algorithm [69] makes an appropriate compromise between computational efficiency and accuracy of results through a new objective function EIOS and a two-level filtering strategy for candidate nodes. In which EIOS replaces the repeated Monte-Carlo simulation, thus reducing the substantial running time and ensuring the accuracy of the results. And a large number of unnecessary calculations are avoided by the two-stage filtering strategy of candidate nodes.

- **Diffusion Model Reduction**. The approach of influence ranking is to effectively estimate influence propagation by converting influence propagation into simpler questions (e.g., PageRank). This approach minimizes processing cost, but the transformed problem may be unrelated to the IM problem, meaning that it might significantly differ from the influence propagation that would actually occur under the diffusion model. Because the node influence ranking ignores diffusion model properties, diffusion model simplification can be used to alleviate this problem. Different diffusion models (IC and LT models) will be introduced in the following sections.

  (a) **IC model**. To activate another user $v$, a user $u$ in the IC model may follow a sizable number of pathways with various activation probabilities. And for model simplification, [71] first proposed the shortest path model (SPM) and its improvement, the SP1M models, considering only significant paths. The idea of SPM is to consider only the shortest path from user $u$ to user $v$. Suppose $d(u,v)$ is the distance between $u$ and $v$. Then the shortest path between any user in the seed set S to user $v$ is $d(S,v) = \min_{u\in S} d(u,v)$, at which time the seed set S has only one chance to be in $d(S,v)$ step to activate user $v$. An improvement of SP1M is that it is possible to activate v at both $d(S,v)$ and $d(S,v)+1$ steps, so that the propagation influence $\sigma$ of a given seed set $S$ can be computed by the Dijkstra shortest path algorithm. Due to the fact that SPM and SP1M only assess the length of the influence paths and disregard the influence probabilities, changes in the influence probabilities of the edges can result in dramatic alterations to the outcomes.
  The best model simplification in the IC model should be the one named maximum influence arborescence (MIA) [18], the main idea behind MIA is to limit a node's influence to a local tree structure rooted at that node. The tree structure can then be used to quickly estimate how that node's influence is spreading. They also extended MIA to PMIA. PMIA does this by updating the tree structure in time after a node is added to the seed set, so that the impact of future seed nodes is not incorrectly estimated by not changing the tree structure. Similar to the idea of MIA is the Independent Path Algorithm (IPA) proposed by [73], with the exception that it also evaluates any paths from node $x$ to node $y$ with propagation probability over the activation threshold.

  (b) **LT model**. With the use of a DAG structure, [18] suggested the LDAG algorithm. It is predicated on the concept that a local DAG structure is built for each node by employing Dijkstra's shortest route method, and the influence of the node is confined to its DAG structure. This is the foundation upon which it is predicated. It does its evaluation of the seeds in a linear fashion and selects the seeds using a greedy hill-climbing strategy. SIMPATH [72] is a scoring-based IM approach that assesses the spread of influence of a seed node or group of nodes by enumerating all simple pathways beginning with the seed node. Unlike LDAG, SIMPATH does not require prior enumeration of all simple paths, but directly estimates the impact distribution on the original graph. Therefore, SIMPATH has a higher time and space efficiency.

  (c) The EASYIM [74] approach, which is applicable to both the IC model and the LT model, estimates the propagation influence of each node by enumerating all influence channels of length. This method is in addition to the methods that have been independently proposed for the IC model and the LT model. It has the advantage of considering the overlap between paths and improving the accuracy.

- **Others**. [75] came up with a computational localization-based RLIM algorithm to solve the problem of choosing the wrong seed nodes because of the "Rich-Club Phenomenon". The nodes of the graph are pruned based on the centrality of the feature vectors to reduce the computational overhead. The computation is then done locally using the localization criteria. Finally, influential nodes are selected by avoiding the rich club phenomenon. [76] designed a new heuristic algorithm CutTheTail (CTT), in order to greatly reduce the memory footprint and expand the computation of the most influential nodes. They were the first to focus on data structures and concepts for space-efficient algorithms, and they discovered a data structure that could be the most space-efficient using a compression framework, thus allowing the processing of large graphs.
- **Summary**. The simulation-based approach is very effective in most cases, and it guarantees a certain approximation ratio. However, the computational overhead required when faced with large social networks with billions of edges is enormous. Heuristic-based methods of either influence ranking or model simplification, taking into account the properties of the diffusion model, are suitable for such scenarios. But, the diffusion model simplification is usually valid only for one model, has poor scalability, and cannot be generalized to other models.

#### 4.2.3. Sampling-based approaches

The purpose of the sampling-based approach is to improve the theoretical efficiency of the simulation-based approach while guaranteeing the approximation ratio. Simulation-based methods estimate the propagation impact of each candidate seed set by running a large number of MC simulations. So to avoid a large number of MC simulations, these methods first perform a large number of samples as well as sampling precomputations in a particular diffusion model and then use these precomputations to estimate the propagation impact. The sampling-based methods are classified into two categories based on the sampling approach: (1) Forward Impact Sampling (FI-Sampling), and (2) Reverse Reachable Sampling (RR-Sampling).

- **Forward Influence**. The core idea of the forward propagation sampling-based approach is to generate a sample by creating instances of the influence diffusion process associated with a defined diffusion model, and then calculate the expected propagation influence $\sigma(S)$ of the seed set S on these samples with approximate guarantees. An example is that, under the ICM diffusion model, the graph $G = (V, E, W)$ constructs a sample $G_i$ by removing each edge $(x, y)$ with probability $\left(1 - p_{x,y}\right)$. Denote by $I_S\left(G_i\right)$ the set of users that the seed set S can influence on the sample $G_i$, and $\left\{G_i, G_2, \ldots, G_m\right\}$ the m samples of the graph G under different instances. Then the expected propagation influence $\sigma(S)$ of the seed set S is the average of the propagation influence of S over these m different samples, i.e., $\sigma(S) = \frac{1}{m}\sum_{i=1}^{i=m} I_S\left(G_i\right)$. Theoretical results show that a greedy framework with a sampling-based approach as an impact estimator can achieve an approximation ratio of $(1 - 1/e - \epsilon)$.
  [64] proposed NewGreedy under the ICM diffusion model. This method generates a number of samples by different instances to evaluate the marginal benefit $(\sigma(S \cup x) - \sigma(S))$ of each node during each iteration. It is worth mentioning that the asymptotic complexity of constructing samples $G_i$ under the IC model is the same as the complexity of running MC simulations. [77] proposed StaticGreedy, which constructs m samples such that the approximation ratio of $(1 - 1/e - \epsilon)$ is guaranteed with probability $\left(1 - n^{-1}\right)$. Although StaticGreedy improves the efficiency, the effect is not obvious in the worst time complexity case. So to further improve the efficiency, [77] also proposed StaticGreedyDU, which uses a pruning strategy to improve efficiency.
  Ohsaka et al. used the index structure combined with the StaticGreedyDU pruning technique to further improve the time efficiency of StaticGreedyDU by proposing PRUNEDMC [78]. It constructs a directed acyclic graph DAG for each sample $G_i$, and each node $x$ in the DAG is a strongly connected component of the sample $G_i$. Then a node with the maximum degree in each DAG is chosen as the central node and the antecedents and descendants of the central node are marked in the corresponding samples. If node $x$ is a precursor of the central node, then PRUNEDMC speeds up the propagation influence estimation of node $x$ by avoiding traversing its successors. Cohen et al. [79] proposed SKIM, which performs a reverse breadth-first traversal (BFS) of the sampling and uses bottom- $K^2$ min- Hash to update the bottom bottom- $K^2$ min-Hash values of the candidate seed set, thus achieving a faster computation of the propagation influence. The worst time complexity of SKIM is similar to that of StaticGreedy. Although the above approach relies on the IC model, it may be extended to other diffusion models (LT, TR, and CT models) because they are all node-independent, i.e., the chance of a user being activated by its neighbors depends only on the neighbors.
  The advantage of Forward Influence is that this method is significantly less time complex compared to Monte Carlo simulation. For Monte Carlo simulation, each time when selecting the seed set S of nodes, it is necessary to calculate the influence of all other candidate nodes after adding the seed set S. And each simulation, it is necessary to conduct at least r rounds, which is very time-consuming. The disadvantage is that it does not perform well in the worst time complexity case and cannot run on graphs with millions of nodes and billions of edges.
- **Reverse Reachable**. Although sampling-based methods can improve the efficiency of simulation-based methods with approximate guarantees. But it does not perform well in the worst time complexity case and cannot operate on graphs with millions of nodes and billions of edges, so Borgs et al. [81] introduced the concept of the reverse reachable set. Let be a forward propagation sample $G'$ constructed on G. Then the reverse reachable (RR) set $\mathcal{RR}_{G'}(v)$ of node v is the set of all nodes on the sample $G'$ that can reach node v. [81] pointed out that by operating on the entire graph. Estimating the sampling influence range by running operations on the whole graph is not necessary. Instead, the propagation influence of the seed set S can be estimated by picking random nodes and seeing how many of those nodes the seed set S can reach. They also proposed a threshold-based method for RIS. RIS generates random RR sets until the number of edges tested hits a preset threshold $\tau$. For better applicability to large-scale social networks, [87] proposed a reverse reachable set-based SRIS algorithm that greatly improves the runtime efficiency without affecting its asymptotic performance. The approach avoids utilizing a preset threshold to choose random RR sets by determining them dynamically during program run.
  To make the RR method more practical and efficient, [80] proposed TIM. TIM enhances RIS by limiting the amount of RR sets while maintaining the approximation ratio. By enhancing the parameter estimation process, they reproposed TIM+ and achieved improved experimental performance with the same level of time complexity. They proposed IMM to increase the performance of TIM/TIM+ even more because it has a better bootstrap parameter estimation technique and is thus more efficient than TIM/TIM+. To ensure theoretical approximation, all of the above methods need to pass a vast group of RR sets, and all of these RR sets need to be preserved in memory, so they need a lot of memory space. To overcome memory limitations, [84] suggested the lazy sampling-based BKRIS method. In order to determine the number of RR sets $m$, BKRIS first utilizes a heuristic to determine a lower constraint on the influence range of the ideal seed set. BKRIS's slow sampling method doubles IMM's speed. [83] proposed an

orthogonal stopping optimization (SSA) method. In order to retrieve the seed set from the currently generated samples, SSA multiplies the number of samples by two for each iteration. The iteration is stopped when the influence of the estimate at the $i$th iteration $\sigma\left(S_i\right)$ is close to the influence of the estimate at the (i-1)th iteration $\sigma\left(S_{i-1}\right)$. They also propose a variant of SSA, D-SSA, and D-SSA ensures an approximation ratio of $(1-1/e-\epsilon)$.

Since the current sampling technique provides effective identification of nodes or groups of nodes with high influence, it is less robust for nodes with low influence. [85] proposed a jump-based influence sampling algorithm (HIS),unlike the general heuristics, the HIS-based method can solve the influence estimation and influence maximization tasks with strict statistical guarantees on the errors. [88] proposed a D-RIS (dynamic reverse reachable set) influence maximization algorithm based on the independent cascade model ICM combined with reverse reachable set sampling. The algorithm first generates the reverse reachable set (RR), and then finds the seed set by using the maximum coverage method to find the k nodes that cover the largest number of RR sets. A randomized modified greedy (RMG) algorithm based on the reverse impact sampling (RIS) technique is proposed for the many times adopted profit maximization (PM2A) problem, which is able to achieve the $(1-1/e-\epsilon)$ approximation ratio with high probability [86].

The advantage of Reverse Reachable is that it does not reuse the extended computational procedure to build the solution to the problem step by step. Instead, it performs all the sampling first to construct the random RR and then selects the entire seed set from it. Thus, this method avoids the "wasteful" expansion computation of the greedy algorithm and most of its successors, since the generated set of random RRs is used to inform the expansion of all nodes in the network. The disadvantage is that the theoretical complexity is higher than that of other IM algorithms.

- **Summary**. In order to avoid re-running MC simulations, a sampling-based approach is proposed, which is divided into two main categories according to the sampling method: Forward Influence and Reverse Reachable.

  For both are different, firstly, from the sampling method, FI is sampled for the whole graph structure, while RR is sampled by randomly selecting nodes in the graph to generate different random RR sets. Then, from the problem solving point of view, FI solves the problem of estimating the propagation influence of a given seed set S and then generates the seed set S using a greedy framework, while RR selects the seed set S by constructing random RR sets with different random nodes. finally, from the performance point of view, it is obvious that RR has much lower time complexity than FI, but the theoretical complexity of RR is also higher than FI.

#### 4.2.4. Meta-heuristic-based approaches

Heuristics are model-specific solutions to a problem, while meta-heuristics are model-independent solutions to a problem. It has been demonstrated that meta-heuristics can be an effective method for tackling NP-hard problems. Therefore, many metaheuristics have been recently considered for solving IM problems, and these methods include simulated annealing (SA) [89,98], genetic algorithms (GA) [90,99,100], discrete particle swarm optimization (DPSO) [101], and colony optimization (ACO) [93], and memory algorithms (MA) [91,92].

- **Simulated Annealing (SA) algorithm**. Simulated annealing simulates metal annealing and optimizes NP-hard problems. [89] used a stochastic-based simulated annealing (SA) algorithm to find the best subset of influential nodes. It proposes to create a metric called expected diffusion value (EDV) to measure how influential candidate solutions are. EDV replaces the traditional time-consuming MC simulation and greatly reduces the computational cost. To address the IM issue in signature networks, [98] used SA. However, due to the influence of the broad search area and parameter values, the SA algorithm frequently succumbs to local optimization. [101] said that effective priority rules can be set, and that tuning parameters can make the metaheuristic algorithm work better.

  [96] suggested an MCDM-based metaheuristic method with three main steps: ranking the nodes, choosing candidates from a pool, and coming up with solutions and judging how good they are. To begin, an MCDM method known as Simple Additive Weighting (SAW) is employed in conjunction with four well-established centrality measures to rank nodes. Second, in the initial level, a user-controlled candidate selection process is presented to choose applicants by excluding less influential nodes. Finally, an adaptive temperature control mechanism is added to the original SA algorithm, proposing that adaptive simulated annealing (ASA) be utilized to develop and evaluate viable solutions.
- **GNA**. GNA combines the NewGreedy algorithm and the genetic algorithm. The main idea is that the NewGreedy algorithm is used to figure out how much each node affects the algorithm.

  [99] used a straightforward evolutionary algorithm to find an approximation solution to the effect maximization problem, and they demonstrated that this method was faster than the greedy approach at finding an exact solution. In order to increase the effectiveness of the impact maximization problem, [90] employed evolutionary algorithms to develop solutions and greedy algorithms to enhance the results. [100] maintained solution diversity by competition and evolution amongst populations, and the outcomes are comparable to the greedy algorithm. [102] proposed a dynamic generalized genetic algorithm to obtain dynamic seed sets in a social network under an independent cascade model. They model several graphs obtained at specified timestamps, which have edges and nodes at different time intervals. Finally, an attempt is made in each graph to locate significant individuals and maximize the influence of people in the social network, which may result in a change in the seed set.
- **Others**. [91] suggested a particle swarm optimization-based DPSO approach to maximize local impact. The discretization rule is used to update the algorithm's generated solution, with the desired local impact estimate serving as the cost function. To address the situation where the EDV function is only applicable to IC models with small activation probabilities, to achieve an optimal balance of runtime and accuracy, the DPSO algorithm creates a new function known as the local impact estimator (LIE). [92] proposed a DDSE algorithm for IMP based on the memory algorithm. They provided a degraded search strategy (DDS) for producing a set of nodes with an influence distribution similar to that of the height method in order to address the issue that employing the predicted diffusion value function results in identifying influential nodes with minimal running time. Then, a genetic algorithm is developed based on the DDS to improve the efficiency. [93] suggested a DBA algorithm based on how smart a group of bats is as a whole. The technique uses greedy local search and discrete rules depending on the network architecture to construct and update populations.

  [94] proposed an adaptive discrete particle swarm optimization algorithm called ADPSO. First, the LP algorithm finds out what the community structure is. Then, the dynamic encoding mechanism of each particle and the discrete evolution rule of the swarm are broken down based on the network community structure to find out how many influential nodes are in each community. To make the seed nodes grow in a reasonable way, a local impact preference strategy is proposed to distribute the number of candidate nodes based on how much each community stands to gain. [95] proposed the GWIM algorithm based on the grey wolf algorithm of IMP. This algorithm varies from earlier metaheuristic algorithms in several ways, including the use of local search to

**Table 6**
A Comparison of the existing context-aware influence maximization algorithms under context-aware diffusion models.

| Context-aware | Categories | | Algorithm | Year | Diffusion model | Approximation | Comparable algorithm |
|---|---|---|---|---|---|---|---|
| Location | OLTM | Simulation-based | IPTB/CT [103] | 2020 | PTC | $1-1/e$ | CELF, IMM, TPHSD & Degree |
| | | Heuristic-based | Expansion/Hint [104] | 2014 | MIA/PMIA | $\epsilon(1-1/e)wrt.MIAstructure$ | PMIA, IRIE, Assembly & Bound |
| | | | TPH [105] | 2015 | TP | $N.A$ | MaxDegree, DD & PageRank |
| | | | MIA-DA/RIS-DA [27,106] | 2016 | MIA/PMIA | $(1-1/e)wrt.MIAstructure$ | PMIA, |
| | | | LTIBM-H [107] | 2019 | MIA | $N.A$ | Greedy, Degree-I, DD-I & Proximity-I |
| | | Sampling-based | Target-IM/Target-IM+ [108] | 2016 | IC-L | $1-1/e-\epsilon$ | MIA-L, Expansion & IMM |
| | | | HIM [109,110] | 2020 | ICM | $N.A$ | WRIS |
| | | | MSTIM [111] | 2021 | ICM | $1-1/e-\epsilon$ | RIS |
| | OBIM | Cluster-based | PartSel [112] | 2018 | MI | $1-1/e$ | TrafficVol, GreedySel & LazyProbe |
| | | | LazyProbe [35] | 2020 | MI | $1-1/e$ | TrafficVol, GreedySel, & PartSel |
| | | Branch-and-bound | expansion-based [36] | 2019 | LI | $\frac{\theta}{2}(1-1/e-\epsilon)$ | CELF & Greedy Cluster |
| | | Greedy | MROAM [113] | 2021 | OI | $N.A$ | G-Order, G-Global, ALS & BLS |
| Time | Discrete time-aware | Heuristic-based | MIA-M/MIA-C [37] | 2012 | IC-M | $(1-1/e)wrt.MIAstructure$ | CELF, MIA & Degree |
| | | | MISP [39] | 2013 | LAIC | $(1-1/e)wrt.MIAstructure$ | MC, ISP, DC, PMIA & MIA-M |
| | | | CT-IPA [38] | 2014 | CT-IC | $(1-1/e)wrt.MIAstructure$ | Random, MaxDegree & Greedy |
| | Continuous Time-aware | Markov Chain | FASTMARGIN [43] | 2015 | DynaDiffuse | $N.A$ | CELF & CELF-IC |
| | | | INFLUMAX [114] | 2016 | TM | $1-1/e$ | IC, SP1M & PMIA |
| | | Sampling-based | IMM [82] | 2015 | CT | $1-1/e-\epsilon$ | TIM, TIM+, SIMPATH |
| Topical | Topic relevant targets | Simulation-based | IMIP/IMAX [115] | 2014 | Expectation | $1-1/e$ | CELF, PMIA, IRIE,CD |
| | | Heuristic-based | MFIP [116] | 2020 | IC LT | $N.A$ | Greedy, DD, TIM |
| | | | LGA/ELGA [117] | 2013 | IC | $N.A$ | LDegree, LRandom, LND |
| | | Sampling-based | CTVM [118] | 2016 | IC | $1-1/e-\epsilon$ | TIM, CELF++, SIMPATH |
| | | | KB-TIM [119] | 2015 | IC | $1-1/e-\epsilon$ | RR, IRR, WRIS |
| | Topic-dependent diffusion | Simulation-based | AIR-Greedy [120] | 2013 | IC LT | $1-1/e-\phi$ | TIC |
| | | Heuristic-based | INFLEX [121] | 2014 | IC | $N.A$ | exactKNN, approxKNN |
| | | | TIM/MIA [122] | 2015 | IC | $\epsilon(1-1/e)wrt.MIAstructure$ | PMIA, INFLEX, MIS |
| | | | MIS/BTS [123] | 2016 | IC | $N.A$ | TA-PMIA, Greedy, PageRank |
| | | Sampling-based | C2IM [60] | 2019 | IC LT | $N.A$ | Random, MaxDegree, CIM |

create neighborhood solutions, the use of discretization criteria, and the use of EDV or LIE as cost functions. This algorithm does not use local search using a specific cost function, as well as using continuous variables instead of directly using graph nodes.

- **Summary**. The main advantage of using metaheuristics is that they can effectively replace time-consuming MC simulations and improve the efficiency of problem solving. Although relatively little research has been devoted to the use of metaheuristics for solving IM problems in this area, it opens up new ways of solving IM problems.

## 5. Context-aware influence maximization

The context-aware IM problem is a variant of the IM problem. In Chapter 2, Section 2, we mentioned that the classical diffusion model only considers the activation probability as a factor in formulating the influence propagation rules. However, the activation between nodes does not simply depend on the activation probability between two nodes, and the seed set found in this way does not have certain validity. Therefore, it is necessary to consider the user's geographical location, the time of information dissemination, the topic of information dissemination, the user's interest, the group effect, product competition, etc. in the dissemination process. Existing research, for context-aware IM problems not only proposes more appropriate diffusion models but also proposes algorithms to solve IM problems under these models. This chapter focuses on three major context-aware IM problems, namely, location, time, and topic. Table 6 summarizes the analysis in this section, and the details are analyzed in the following subsections.

### 5.1. Location

The location-aware influence maximization (LAIM) problem, which combines the traditional IM problem with spatial indices, has been a relatively popular area of research in recent years due to the proliferation of GPS-equipped smart devices (such as smartphones) and the popularity of location-based social networks (such as Foursquare, Twitter, etc.). The purpose of LAIM is to maximize the influence of those specified users who are related to location requirements, rather than all other users in the social network. The direction of location-aware IM based on its application scenarios can be roughly divided into two categories [15], online location-aware influence maximization (OLIM) and offline billboard influence maximization (OBIM). The key difference between OLIM and OBIM problems is the influence diffusion process. In the OLIM problem, users' influence can be diffused through online social networks. But in the OBIM problem, a user can be influenced by an outdoor billboard only when he passes the billboard within a certain range.

#### 5.1.1. Online location-aware IM (OLIM)

The online location-aware influence maximization (OLIM) problem is a variation on the traditional influence maximization (IM) problem. It looks at how the location of target users in an online social network

affects the spread of their influence. Unlike the traditional IM problem, the goal of both is to find a set of seed users $S$ whose influence in an online social network is as large as possible. However, they differ in the definition of target users and the influence diffusion process. the target users in the IM problem are any users in the social network. the target users in OLIM may need to satisfy location constraints, e.g., the distance between the seed users and the target users does not exceed a specific threshold value. In addition, the influence diffusion process for evaluating a particular seed user in the OLIM problem also needs to consider the influence of the user's location. The research work in the OLIM problem dedicated to overcoming the theoretical difficulty of estimating the impact diffusion $\sigma(S)$ of the seed set S is as follows.

- **Simulation-based Approach**. [103] pointed out that existing studies only consider the use of geographic distance to study influence maximization and ignore the impact of user spatio-temporal behavior on information dissemination or location enhancement. So they considered the impact of user spatio-temporal behavior and proposed a similarity-aware influence maximization (SIM) problem, which aims to maximize influence propagation by effectively measuring the impact of user spatio-temporal behavior. In addition, they propose two greedy algorithms, the influence propagation tree (IPTB) based algorithm and the cut tail (CT) algorithm, to maximize the influence propagation. In the IPTB algorithm, not only an effective strategy to prune irrelevant nodes, but also a method to calculate the upper limit of user influence is proposed. In the CT algorithm, irrelevant nodes are pruned based on NC, and submodular attributes are used to find the most influential nodes.
- **Heuristic-based Approach**. [104] focused on region queries. They defined the IM problem as finding k users as a seed set S in a given target geographic region R such that these k users influence the maximum number of users in the target region R, i.e., maximize the propagation impact on the region R. This work is based on the standard IC model, which uses the MIA/PMIA model [18] to estimate the propagation impact of the seed set. [104] made the following three efforts to solve the LAIM problem: Firstly, the Expansion algorithm is proposed, which designs an optimal priority search framework that prioritizes access to users with a large influence ceiling and prunes those without excessive influence. Second, boundary estimation techniques for pruning have been developed to perform better pruning operations. To rapidly discover the users who have a significant impact on $R$, they employ the traditional spatial index QuadTree. In order to accomplish effective upper and lower limit estimation, a hint-based technique is suggested that first computes seed sets in advance for each leaf region of the QuadTree. These seed sets are then used as suggestions for regions intersecting with R.
However, users who are outside of the intended zone but only a short distance away are ignored in [104]. As a result, some studies [27,106,108] took into account the user's distance from the event place rather than the region. The objective is to increase the number of users who have activated within a short distance of the place. [27,106] proposed a pruning-based framework similar to [104], but they employed a method known as the distance-aware pruning-based algorithm MIA-DA to concentrate on optimizing the influence of potential customers at a specific promotion location. The algorithm uses the user's distance from the query location as edge weights, using the ICM diffusion model under the MIA/PMIA framework [18], which makes the influence propagation maximized. For boundary estimation, they choose a collection of anchor locations that keep the influence of each user in the same way. At query time, they use triangular inequalities to figure out the boundary from the anchor locations.
To study the problem of product promotion in the O2O model, [105] suggested a TPH model, which is an improved influence diffusion model based on the IC model that may effectively depict the product adoption process in an O2O setting. The model has a process for how information spreads based on where the event is and how likely it is that users will be there. The effect of each node in the model is then measured using a centrality, and the node with the highest centrality is chosen as the seed node. A method for figuring out the chance of being offline has been developed after examining the user's movement behavior. A location-based influence maximization technique called TPH is suggested, taking into account the influence capabilities.
[107] thought about the location-aware target influence blocking maximum (LTIBM) problem. The goal of this problem is to find a set of positive seeds in a given social network that will block as much as possible the influence of negative seeds on target nodes that are in a given region and like a given set of topics. They came up with LTIBM-H, a heuristic algorithm based on QT-tree and maximum impact tree structure, to make the greedy algorithm work better (MIA).
- **Sampling-based Approach**. [108] proposed Target-IM/Target-IM+, which is the same topic studied by Wang et al. [27,106]. But they used the reverse reachable sampling (RR-Subgraph) approach to estimate the propagation influence of the seed set instead of the MIA/PMIA framework used by [27,104,106]. And they guaranteed the $(1-1/e-\epsilon)$ approximation ratio. [109,110] focus on the overall impact maximization (HIM) query problem. They proposed an overall impact diffusion model, HIM, based on an IC model considering actual user interactions on the Web and offline. To solve the HIM query, they proposed a basic algorithm that uses the inverse k-nearest neighbor technique to evaluate the offline impact diffusion and a sampling-based approach to compute the propagated impact under the HIM model. In addition, they developed an index-based method to handle large networks. [111] focused on the multidimensional selection-based target impact maximization problem MSTIM, and proposed an index-based approach to the MSTIM problem, which utilizes the idea of reusing previous results and exploits the coverage relationship between queries to achieve an efficient solution of MSTIM.

#### 5.1.2. *Offline billboards IM (OBIM)*

Offline billboard influence maximization (OBIM) problem arises due to the emergence of trajectory data of moving objects and information about offline facilities. The OBIM problem is quite different from the traditional IM problem. In terms of problem definition: given a trajectory database $\mathcal{T}$, a billboard $\mathcal{B}$ where the advertisement is placed and a budget $L$, the OBIM problem aims to find a set of billboard placement locations $S \subset U$ that maximizes the affected trajectory. Secondly, there is also a great difference in the use of diffusion models. The diffusion models widely used in the OBIM problem are, respectively, One-time impression (OI) model [34], Multi-time impression (MI) model [35], and the Logistic influence (LI) model [36]. Regarding the solution of the OBIM problem, specific algorithms are divided into three categories: (1) Cluster-based approach, (2) Branch-and-bound approach, and (3) Greedy algorithm.

- **Cluster-based Approach**. [35,112] focused on the study of individual advertisers achieving maximum impact by selecting an optimal set of billboards under budget constraints. They pointed out that there is an overlap in the impact of billboards in different regions on trajectories. The cluster-based approach exploits the localized nature of billboard influence to avoid the overlap of influence. That is, the billboards are first divided into a set of small clusters with low impact overlap, and then the local solution of each cluster is found by enumeration. Finally, the global solutions are constructed based on the local solutions of each cluster. In addition, they improve the time efficiency of the algorithm

by pruning the billboards with low benefit or cost ratios. [124] also used a cluster-based approach by dividing the trajectory database into clusters and accessing the clusters sequentially, and also proposed a trajectory metric to reduce the cost of estimating impacts. However, unlike [35,112,124] did not consider impact diffusion.

- **Branch-and-bound approach**. [36] focused on the outdoor advertising display count optimization (ICOA) problem, and the purpose of the study was the same as [112], but [36] used the LI model to evaluate the impact propagation of billboards. And the objective function of the ICOA problem is non-submodular under the LI model, so there is no effective algorithm to maintain a constant approximation rate. To overcome this challenge, they propose an estimation method that computes an upper bound by using dynamic tangents to tightly constrain the actual influence.
- **Greedy algorithm**. [113] investigated the OBIM problem from the perspective of the moderator responsible for assigning billboards to all advertisers, and formulated it as a minimization regret for the out-of-home advertising market problem (MROAM). The MROAM problem aims to minimize the regret of influencer providers when dealing with a large number of influencer buyers. An OI model is first used to evaluate the spread of influence between billboards and users, and then two greedy heuristics are proposed as baselines to solve the MROSAM query. Since the objective function of MROSAM is non-submodular and non-monotonic, the greedy methods may produce poor local minima, so the authors further propose a stochastic local search framework to avoid local minima.

### 5.2. Time

The downside of the discrete-time diffusion model in the conventional IM problem is that the diffusion process comes to an end when all of the nodes have been activated. But it can take a while for the influence dissemination process to come to an end. The diffusion process must therefore adhere to a time limit in order to solve the time-aware influence maximization (TimeAIM) problem. The section on diffusion models has provided a thorough description of the diffusion model utilized there.

#### 5.2.1. Discrete time-aware IM (DTIM)

Some IM methods [37–40] dedicated to the study of discrete-time perception are presented below.

[37] introduced the IC-M model based on the traditional ICM model. They defined the IM problem as maximizing the influence propagation within a given deadline. And to solve the problem of too low efficiency of the approximation algorithm, two MIA-based heuristics are designed: the first one is based on the dynamic programming algorithm MIA-M to compute the exact spread of influence in the tree structure, while the second one is the MIA with transformed propagation probability, i.e.MIA-C, which first estimates a transformed propagation probability by combining the satisfaction probability, the influence probability and the final time constraint $\tau$ to estimate a transformed propagation probability designed to simulate the impact diffusion under the IC-M model. These transformed probabilities are then used as parameters of the IC model, and the MIA algorithm proposed for IC is run to select the k seeds.

[38] was presented by taking into account a new impact diffusion model known as the continuous activation and time-constrained IC (CT-IC). Additionally, a scalable independent path algorithm (CT-IPA), which expands the scalable independent path algorithm IPA of the IC model, is provided for optimizing impact under the CT-IC paradigm.

[39,40] suggested a delay-aware independent cascade (LAIC) model, and three approximate techniques, including a simulation-based approach and two impact propagation path-based algorithms (MISP), are offered for solving the IM problem. However, the disadvantage of the impact propagation path-based approach is the need to manually adjust the parameter $\theta$ so as to maintain a good balance between influence propagation and runtime.

#### 5.2.2. Continuous time-aware IM (CTIM)

The continuous-time independent cascade model was first proposed by [41]. The next year, they proposed an improved and efficient randomized estimation algorithm, CONTINEST, and an impact maximization algorithm, INFLUMAX [114]. The CONTINEST algorithm provides approximate impact estimation with provable guarantees, and the INFLUMAX algorithm efficiently finds the approximate optimal set of source nodes with provable guarantees by using submodularity and can use any of the estimation algorithms as a subroutine of the influence estimation. Therefore, using CONTINEST as a subroutine of INFLUMAX enables efficient influence maximization and can be extended to large networks of millions of nodes.

[43] proposed the DynaDiffuse diffusion model, which relies on stochastic model checking, for estimating the impact of an initial set of nodes. Based on this, a fast greedy algorithm called FASTMARGIN is designed for finding the optimal seed set. The algorithm first uses stochastic model checking with parallel combinations of CTMC to predict a relatively accurate influence diffusion, and then when selecting additional nodes to be added to the current initial set, for each node $v \in V$, the marginal benefit influence diffusion $\left(\sigma_T(I \cup \{v\}) - \sigma_T(I)\right)$. For efficiency, they used the discount formula to estimate this marginal impact diffusion instead of determining the exact value by stochastic model checking.

### 5.3. Topic

The combination of the topic about which information is being disseminated and the traditional IM problem is called the Topic-Aware Influence Maximization (TAIM) problem. The TAIM problem considers influence propagation to be dependent not just on the specified seed set $S$, but also on topics, where topics reflect item qualities and user interests. Given a topic as a query, it seeks the optimum seed set for maximizing topic-perceived impact. Existing studies are divided into two categories: (1) TAIM with topic-related goals, where the users' own topic interests are considered, with the aim of maximizing the impact on a subset of users related to the query topic. (2) Topic diffusion IM, which considers the topic of the information disseminated among users, i.e., the topic of the information.

#### 5.3.1. Topic relevant targets

While TAIM for topic-related targets focuses on the characteristics of the users themselves, the following studies introduce the concept of gain to distinguish users by considering the expected sum of activation user gains as propagation influence $\sigma(x)$. The IM problem is redefined as an influence maximization problem under a gain calculation model.

[119] proposed a topic-aware query model KB-TIM that considers how users match query topics to compute user benefits. This model filters the target users based on their personal profiles. The user's preferences for particular topics are detailed in their profile, e.g., the user profile $\{\langle smarthome, 0.7\rangle, \langle car, 0.3\rangle\}$. Then, given a query topic, e.g., smart home, the user's interest is 0.7 because only the smart home topic matches the query. They used the ICM diffusion model and the reverse reachable sampling (RR) algorithm to solve the IM problem.

[118] generalized the TAIM problem by using a predefined target impact function. They also used reverse reachable (RR) sampling, but they utilized the period termination rule to avoid generating too many RR sets. [117] presented a particular example of the TAIM problem. The problem aims to find those users who are relevant only to the query topic and not to other users. That is, the affected users are only affected by the topic and not by other users. They presented two ICM model-based algorithms called Local Greedy (LGA) and Efficient Local Greedy (ELGA). By adapting some trimming algorithms to the local structure of the target user, the LGA is a simulation-based method that increases query efficiency. ELGA is a heuristic method that ensures the shortest path between each user and the target user.

[116] proposed a multi-factor-based information dissemination model (MFIP) by extending the classical information dissemination model by considering various factors, such as information content and user characteristics. This paradigm divides the spread of information into two phases: the receiving phase and the acceptance phase. And they proposed an efficient heuristic algorithm, namely weighted degree reduction (WDD), to select the seed nodes under the MFIP model.

#### 5.3.2. Topic-dependent diffusion

The TAIM of topic-related diffusion focuses on the topic of the disseminated information. The following studies consider the edge $(x, y)$ between a pair of individuals $x$ and $y$ to be topic-related, i.e., individual $y$ may be activated by individual $x$ on some topics (e.g., smart homes) but not on others (e.g., cars).

[120,121] proposed the topic-dependent TAIM problem, noting that similar searches will have similar impact ranges. Moreover, they proposed an indexing technique (INFLEX) that makes use of similarity search and precomputation of seed sets. The seed set under each topic distribution is first precomputed by an arbitrary IM method using samples from a selection of topic distributions. [123] provided solutions for some special graphs that follow some properties such as subadditivity, typical separability, etc., based on the framework proposed by [121]. However, the above methods do not provide theoretical approximation guarantees, so [122] improved the methods of [120,121,123] by adopting the MIA/PMIA framework under the ICM model. To estimate an upper bound on the propagation influence $\sigma(x)$ of user $x$, they propose a best-effort framework. The propagation influence of the seed set $S$ is then estimated based on the upper limit of influence and pruned for unimportant users.

An interest-based parallel social behavior algorithm is proposed in order to maximize the influence in social networks. The algorithm [125] dynamically weights the user's interactions into social behaviors along with the user's interests, and computes the optimal set of influencing nodes by implementing a machine with a CPU architecture with perfect parallelism through the community structure.

## 6. Research challenges and future directions

### 6.1. Challenges

- **Determining the Incorporation Stability of IM Algorithms.** [126] pointed out that when the input influence probabilities have adversarial noise, the IM algorithm is less stable. In other words, a tiny modification in the diffusion model's influence probability could have a significant impact on the optimal subset. Although there exist some studies that specifically address this issue, such as IRIE and EASYIM, they all assume that the social graph structure is constant. However, in practice, the graph structure is dynamic. Therefore, it can be difficult to derive a reliable seed set from a given dynamic social graph.
- **No longer restricted to strict submodularity.** Because the objective influence function is submodular, there is a solution to the IM problem. However, in some circumstances, the target function's submodularity property is overly stringent. If the target influence function is not submodular, then the greedy framework cannot be used, and thus a good theoretical approximation ratio cannot be guaranteed. So modeling for more general functions is more in line with practical needs. For example, the weak submodular function proposed in [127].
- **Considering group norms.** The current IM algorithm basically focuses only on the influence between two nodes. But in real life, people are greatly influenced by group effects, such as herding behavior. We can be influenced by the behaviors of people with similar educational backgrounds, experience, and age groups around us. For example, [128,129] presented seminal work on the consistency-perception IM problem, but they ignored the characteristics of users regarding social groups.

### 6.2. Directions

- **Designing algorithms for specific datasets.** Different social network datasets have different properties and characteristics, and existing IM methods are used to find suitable datasets from the perspective of problem solving. But it would be convenient to design methods to solve IM problems from the characteristics of datasets. To the best of our knowledge, no research has ventured into this area.
- **Enables information diffusion models to gain realistic complexity.** The traditional diffusion model is a simplification of the complexity of the real-life diffusion process. A user's state changes depending on the relationship between users and the influence of social, environmental and other neglected factors. Therefore, in order to cope with the complexity of reality, it is important to consider the social environment and the intrinsic ideological preferences of users.
- **Using deep learning techniques to solve IM problems.** Machine learning and deep learning techniques are making a splash in other fields, so using these techniques to solve IM problems is an inevitable trend for the future. These techniques are not only used to build diffusion models but can also be used in many extended scenarios of IM problems, such as the exploration of unknown social networks, etc.

## 7. Conclusions

This review provides a refinement classification of the existing advanced IM algorithms from an algorithmic perspective. First, we briefly explain the efforts made in traditional diffusion models and in building diffusion models using deep learning techniques. Then, the traditional IM problem is defined, and the theoretical difficulties and feasible solutions to solve the IM problem are analyzed. Next, we present a rigorous theoretical comparison study between traditional IM algorithms and context-aware IM algorithms. Finally, future research directions and challenges are also pointed out for the IM problem. Our review provides the latest developments in the industry and has a better understanding of IM problems for researchers who are new to the IM direction and provides a good starting point for work in the field.

## CRediT authorship contribution statement

**Yuxin Ye:** Literature collection and classification summary, article framing and writing. **Yunliang Chen:** Literature summarization, model performance comparison. **Wei Han:** Article format checking, paper typesetting.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yunliang Chen reports financial support was provided by National Natural Science Foundation of China.

## Data availability

No data was used for the research described in the article.

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